

Data Science 6

Mark

Lecture 1

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Introduction

W.E Du Bois graphs post encmapaction. Data can explore, infer, and predict.

1 Intro

1.1 Exploration

Abstraction (taking large data and abstracting it to simplify) and Algorithmic thinking (creating systems to process data). Mostly what are we doing in this class. Ex: graphs from berkeley during covid

Bias in depixlation graphs, based on bias traning data (white features) and also hallucinations. Look at data6.org and datahub.berkeley.edu for hw and lecs and gradescope.

2 What are computer programs

What is python: basically a lanauge that is simplefed then by the computer to run. Using jupyter notebooks instead of ides. We will use datahub.berkeley.edu on a berkeley hosted server that runs jupyter notebooks.

2.1 Arithmetic

Expressions are evaluated from left to right

- Expressions have values (2) and operators (like + and **)
- Expressions evaluate to values

Ex: $\text{pow}(2, 3) \rightarrow 8$

Syntax rules are like PEMDAS for math, python uses PEMDAS to eval math expressions. Recall the basic operators.

Operator	Meaning	Example	Result
+	Addition	5 + 3	8
-	Subtraction	10 - 4	6
*	Multiplication	6 * 2	12
/	Division	8 / 2	4.0
//	Floor Division	9 // 2	4
%	Modulus (Remainder)	10 % 3	1
**	Exponentiation (Power)	2 ** 3	8

Figure 1: Common Python Operators

Note our number system is base 10. So modulating a number by 10 gives the last digit and flooring the number by 10 gives everything but the last digit.

Note. Shift + enter to run quickly. Also only last evaluated command is outputted.

 Edit vs command mode

USE COMMENTS to tell you what the code acutally does. Good practice.

2.2 Data Types

Do diffrent things and have diffrent limts, consider when to use them and the benifits.

- ints
- floats
- booleans
- strings

Note. REMEMBER Exps return ints when use only ints unless a float is used

Quick quiz — check your understanding

1. What are the two core skills for working with data that this class emphasises, and what does each mean?
2. Give two reasons the depixelation / image models produced biased results.
3. In what order does Python evaluate an expression, and what rule governs the precedence of the math operators?
4. What does `pow(2, 3)` evaluate to? Which operator gives the same result?
5. Our numbers are base 10. What does `n % 10` give you? What does `n // 10` (floor) give you?
6. In a Jupyter notebook: which keystroke runs a cell quickly, and how many of the cell's expressions get displayed as output?
7. List the four basic data types introduced.
8. If an expression uses only `ints`, what type does it return — and what one thing changes that?

Answers

1. *Abstraction* (simplifying large data down to what matters) and *algorithmic thinking* (building systems/steps to process data).
2. Biased training data (features skewed toward white faces) and model *hallucination* — it invents plausible-but-wrong detail.
3. Left to right, following **PEMDAS** for operator precedence.
4. `pow(2, 3) = 8`; the equivalent operator is `**` (i.e. `2 ** 3`).
5. `n % 10` gives the *last digit*; `n // 10` gives *everything except the last digit*.
6. **Shift + Enter**; only the *last* evaluated expression in the cell is shown.
7. `int`, `float`, `boolean`, `string`.
8. It returns an `int` — unless a `float` appears anywhere in the expression, which makes the result a `float`.